

q, t -Catalan Middle Coefficients

1 Classical flat middle coefficients

Let

$$C_n(q, t) = \sum_D q^{\text{area}(D)} t^{\text{dinv}(D)}$$

where the sum is over Dyck area sequences D of length n . Put

$$M = \binom{n}{2}, \quad \text{defc}(D) = M - \text{area}(D) - \text{dinv}(D).$$

The Dyck-skeleton decomposition formula has the following flat-middle consequence.

Theorem 1 (Flat middle coefficients in the skeleton range). *For $n \geq 4$, $M = \binom{n}{2}$, and $0 \leq d \leq 2n - 8$, the coefficient of*

$$q^j t^{M-d-j}$$

in $C_n(q, t)$ is independent of j for all integers

$$d \leq j \leq M - 2d.$$

The common value is the number of special Dyck skeletons of length n and deficit d .

Here a special Dyck skeleton is a full Dyck skeleton other than the exceptional full skeleton

$$(0, 0, 1, 0, \dots, 0, 1)$$

when that exceptional sequence exists. The input from the skeleton decomposition is that every special skeleton S of deficit d satisfies $\text{area}(S) \leq d$, and the skeleton contribution therefore fills the displayed middle band with the same coefficient.

2 Broader conjecture

The proved statement above is restricted to $0 \leq d \leq 2n - 8$. The broader flat-middle prediction is the following.

Conjecture 1 (Full flat-middle range). *Let $M = \binom{n}{2}$. For any $n \geq 1$ and*

$$0 \leq d \leq \left\lfloor \frac{M}{3} \right\rfloor,$$

the coefficient of $q^j t^{M-d-j}$ in $C_n(q, t)$ is independent of j for all integers $d \leq j \leq M - 2d$.

This conjecture is not a theorem in this item. The curated code gives bounded checks of the theorem range only.

3 Computer check

The script `code/check_flat_middle_coefficients.py` is a self-contained port of the bounded check from the Dyck repository. It generates all Dyck area sequences for $4 \leq n \leq 8$, computes direct coefficients in the convention $q^{\text{area}} t^{\text{dinv}}$, counts special skeletons by deficit, and verifies the flat band equality for all $0 \leq d \leq 2n - 8$.

The default run checks 2047 Dyck sequences, 25 (n, d) -bands, and 325 individual middle-band coefficients. It also checks that the formula

$$\text{defc}(D) = \binom{n}{2} - \text{area}(D) - \text{dinv}(D)$$

agrees with the explicit deficit-pair count on every generated sequence.

These computations are regression evidence for the implementation and finite evidence for the displayed statements; they do not replace the skeleton decomposition proof and they do not prove the full flat-middle conjecture.